Project Overview

Topic: Neurosymbolic Reasoning

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January 30, 2026

Contents

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Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \mathrm{Human}(x) \rightarrow \mathrm{Wibu}(x)$

(All humans are wibu.)

$\mathrm{Human}(\mathrm{Huy})$

(Huy is human.)

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$$\frac{\begin{array}{l} A \rightarrow B \\ A \end{array}}{\therefore B}$$

(Syllogism/Modus ponens)

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- Explainability
- Data Efficiency
- Abstract Reasoning

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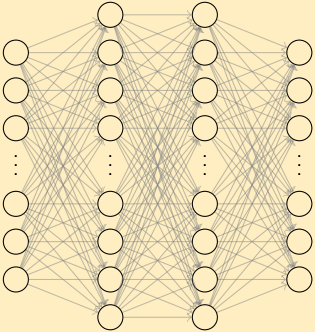
$\Rightarrow$ **reliable but inflexible.**

Modern AI: Data-driven/Neural Networks

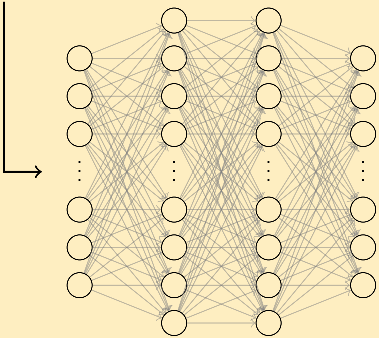
Neural Networks



Modern AI: Data-driven/Neural Networks

Neural Networks

"Is Huy a good boy?"



Modern AI: Data-driven/Neural Networks

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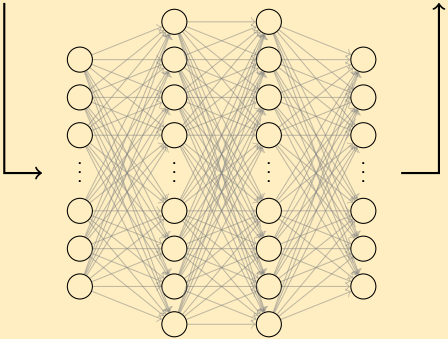

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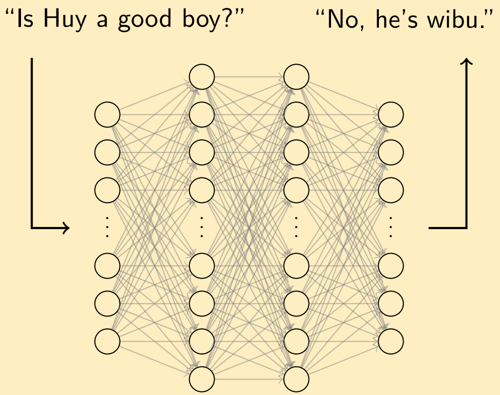

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- Perception
- Automated Feature Extraction

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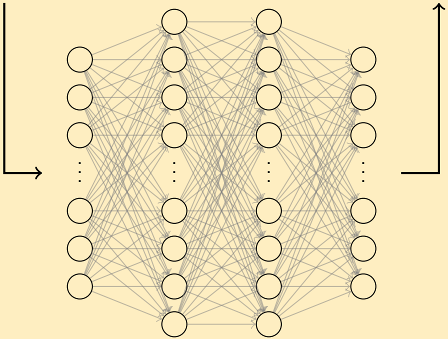

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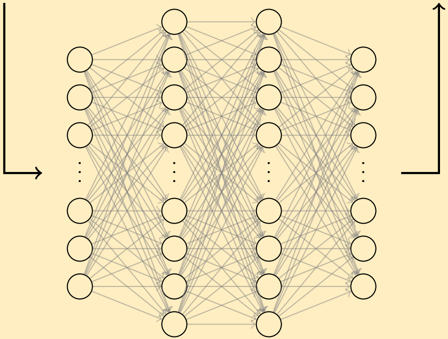

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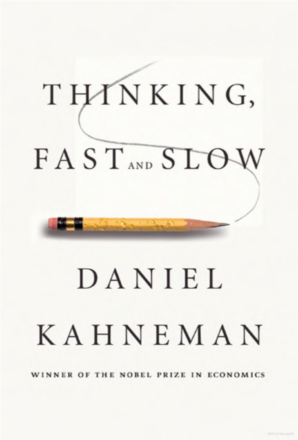
$\Rightarrow$ **flexible but unreliable.**

Neurosymbolic AI: A Hybrid Approach



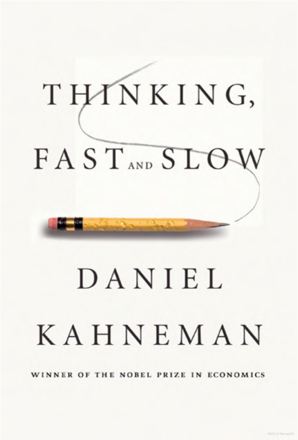
SYSTEM 1 (FAST)

Intuition & Instinct

SYSTEM 2 (SLOW)

Rational thinking

Neurosymbolic AI: A Hybrid Approach



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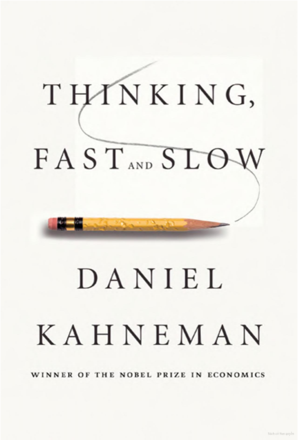

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॥	॥
NEURAL NETWORK	**SYMBOLIC ENGINE**

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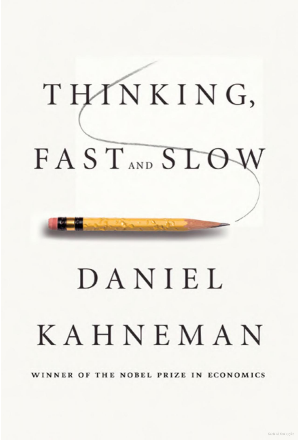

SYSTEM 1 (FAST)		SYSTEM 2 (SLOW)		HUMAN INTELLIGENGE
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||		||		
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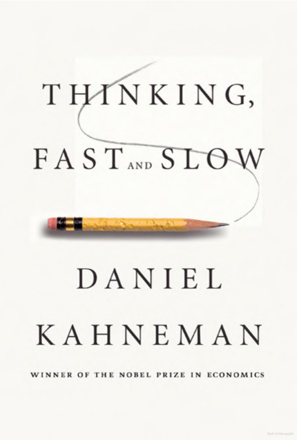

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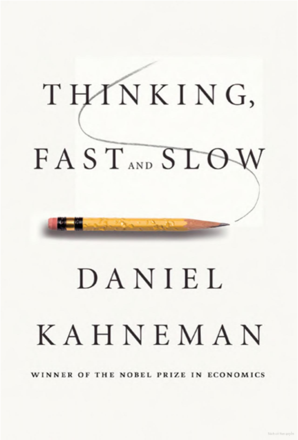

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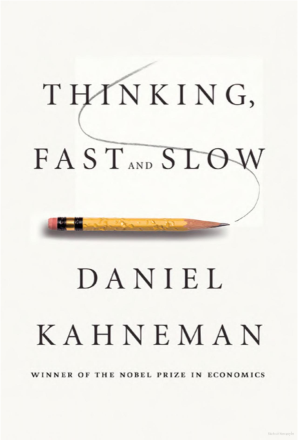

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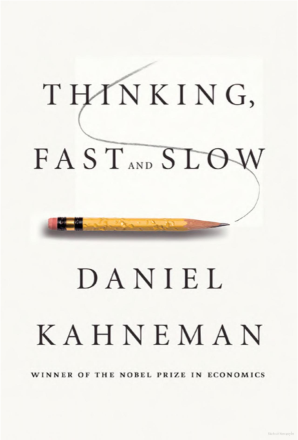

Semantic Parsing/Language Formalization

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NEURAL NETWORK $\oplus$ **SYMBOLIC ENGINE** = **NEUROSYMBOLIC AI**

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Problem Statement

Language Formalization

The goal is to learn a mapping function $f : \mathcal{X} \to \mathcal{Y}$ that translates Natural Language sentences (x_{NL}) into machine-verifiable First-Order Logic formulas (y_{FOL}).

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Input (x_{NL})

Natural Language
(Unstructured)

"Every student takes a course."

$$\xrightarrow{f(\cdot)}$$

Output (y_{FOL})

First-Order Logic
(Structured)

$\forall x(\text{Student}(x) \rightarrow \exists y(\text{Course}(y) \wedge \text{Takes}(x, y)))$

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Core Challenges:

- **The Requirement for Absolute Precision:** Formal languages are syntactically fragile. A formal expression must be 100% correct to be machine-verifiable.
- **Data Scarcity and the Expert-Data Bottleneck:** Creating NL-FOL pairs requires professionals with deep knowledge of both the domain and the formal syntax.

Contents

Data Source

The data used consists of the **training sets** of the datasets listed in the table below.

Source	Volume	Description
FOLIO	1,000 samples	**With FOL label**, with true/false/uncertain label.
ProofWriter	41,400 samples	**Without FOL label**, with true/false label, with proof.
RuleTaker	480,000 samples	**Without FOL label**, with entailment/not entailment label.
ReClor	4,637 samples	**Without FOL label**, in multiple-choice format.
LogiQA	7,380 samples	**Without FOL label**, in multiple choice format.

Table: Preliminary statistics of data sources.

Data Source

FOLIO Dataset

FOLIO Dataset

FOLIO is an expert-written, open-domain, logically complex and diverse dataset for natural language reasoning with First-Order Logic.

Examples:

Premises	Premises-FOL	Conclusion	Conclusion-FOL	Label
All rabbits have fur. Some pets are rabbits.	$\forall x\ (\mathtt{Rabbit}(x) \rightarrow \mathtt{Have}(x, \mathtt{fur}))$ $\exists x\ (\mathtt{Pet}(x) \wedge \mathtt{Rabbit}(x))$	Some pets do not have fur.	$\exists x\ \exists y\ (\mathtt{Pet}(x) \wedge \mathtt{Pet}(y) \wedge \neg\mathtt{Have}(x, \mathtt{fur}) \wedge \neg\mathtt{Have}(y, \mathtt{fur}))$	Uncertain
All tigers are cats. No cats are dogs. All Bengal tigers are tigers. All huskies are dogs. Fido is either a Bengal tiger or a cat.	$\forall x\ (\mathtt{Tiger}(x) \rightarrow \mathtt{Cat}(x))$ $\forall x\ (\mathtt{Cat}(x) \rightarrow \neg\mathtt{Dog}(x))$ $\forall x\ (\mathtt{BengalTiger}(x) \rightarrow \mathtt{Tiger}(x))$ $\forall x\ (\mathtt{Husky}(x) \rightarrow \mathtt{Dog}(x))$ $\mathtt{BengalTiger}(\mathtt{fido}) \oplus \mathtt{Cat}(\mathtt{fido})$	Fido is neither a dog nor a husky.	$\neg\mathtt{Dog}(\mathtt{fido}) \wedge \neg\mathtt{Husky}(\mathtt{fido})$	True
...	...	...	...	...

Table: Sample data from the FOLIO dataset.

Data Source

ProofWriter Dataset

ProofWriter Dataset

ProofWriter is a synthetically generated dataset for logical reasoning over natural language.

Examples:


```
{
  "en": "$answer$ ; $proof$ ; $question$ = Anne is green. ; $context$ = sent1: Anne is kind. sent2: Fiona is round. sent3: Fiona is smart. sent4: Gary is nice. sent5: Gary is smart. sent6: Gary is white. sent7: Harry is kind. sent8: If something is cold and not round then it is smart. sent9: If something is green and white then it is smart. sent10: Cold things are nice. sent11: If something is cold then it is nice. sent12: If Gary is round and Gary is smart then Gary is kind. sent13: Nice things are green. sent14: If Anne is white then Anne is smart. sent15: Kind things are white. sent16: If something is kind and smart then it is cold.",

  "ro": "$answer$ = True ; $proof$ = # sent13%int1 # sent10%int2 # sent16%int3 & sent1 # sent14%int4 # sent15%int5 sent1 ; with int1: Harry is furry. ; int1: Harry is big. ; int1: Harry is smart. ; int1: Harry is quiet."
}
```

Data Source

RuleTaker Dataset

RuleTaker Dataset

RuleTaker is a synthetic dataset for evaluating multi-hop logical inference over explicit facts and rules.

Examples:

Context	Question	Label	Config
Bob is red. Charlie is smart. Fiona is smart. Gary is furry. All blue, white things are rough. White, red things are big. If something is white and smart then it is red. All blue things are smart. Blue, rough things are furry. If something is blue then it is red.	Charlie is smart.	entailment	depth-0
The bear likes the dog. The dog is nice. The dog likes the bear. If someone needs the dog then they like the dog. If someone does not like the bear then they are rough. If someone is rough then they do not like the bear.	The bear is not rough.	not entailment	depth-2
Anne is green. Anne is smart. Anne is young. Dave is big. Dave is young. Gary is green. Gary is kind. Gary is round. Gary is young. Harry is smart. All smart, green things are round. If something is smart then it is kind. All smart, green things are white. All white things are big. If Dave is white and Dave is young then Dave is green. All kind, young things are white. All young things are smart.	Gary is green.	entailment	depth-5
...	...	...	...

Table: Sample data from the RuleTaker dataset.

Data Source

ReClor Dataset

ReClor Dataset

ReClor is a reading comprehension dataset for assessing logical reasoning in LLMs via context-based multiple-choice tasks. 91.22% of it are from actual exams of GMAT and LSAT while others are from high-quality practice exams.

Examples:

Context	Question	Answers	Label
Shipping Clerk: The five specially ordered shipments sent out last week were sent out on Thursday. Last week, all of the shipments that were sent out on Friday consisted entirely of building supplies, and the shipping department then closed for the weekend. Four shipments were sent to Truax Construction last week, only three of which consisted of building supplies.	If the shipping clerk's statements are true, which of the following must also be true?	["At least one of the shipments sent to Truax Construction last week was sent out before Friday.", "At least one of last week's specially ordered shipments did not consist of building supplies.", "At least one of the shipments sent to Truax Construction last week was specially ordered.", "At least one of the shipments sent to Truax Construction was not sent out on Thursday of last week."]	0
...	...	...	...

Table: Sample data from the ReClor dataset.

Data Source

LogiQA Dataset

LogiQA Dataset

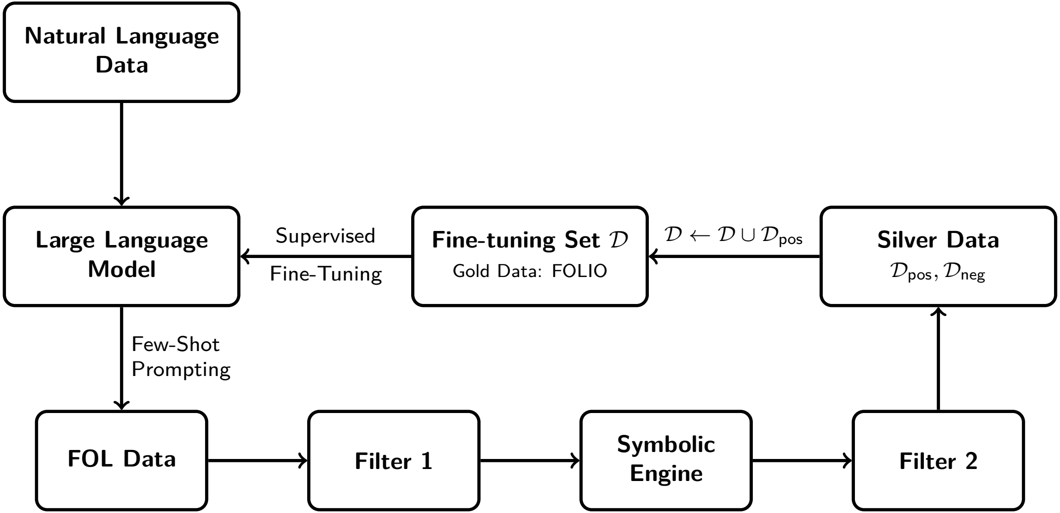
LogiQA is constructed from the logical comprehension problems from publically available questions of the National Civil Servants Examination of China.

Examples:

Context	Query	Options	Correct Option
There was a group discussion of judicial workers in the city. One group has 8 people. At the beginning of the meeting, the group leader asked everyone if they knew each other. As a result, only one person in the group knew 3 of the group, 3 knew 2 of the group, and 4 knew 1 of the group.	If the above statistics are true, which of the following conclusions can best be reached?	["The group leader knows the most in the group, and the others know each other less.", "This is the first time such a meeting has been held and everyone is new.", "Some members may only know what they have seen on television or at a briefing.", "Although there are not many acquaintances in the group, what they knew are all close friends."]	2
. . .	. . .	. . .	. . .

Table: Sample data from the LogiQA dataset.

Data Pipeline



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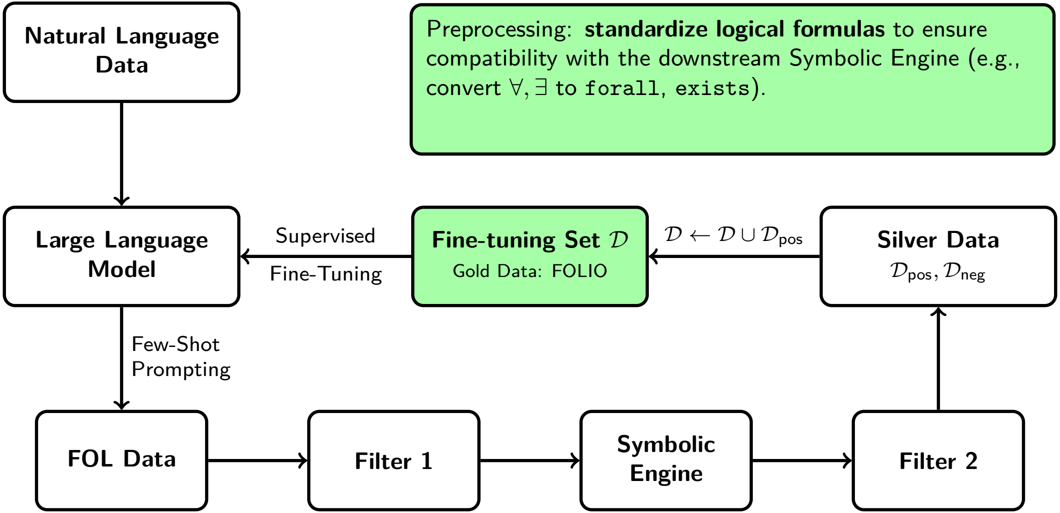
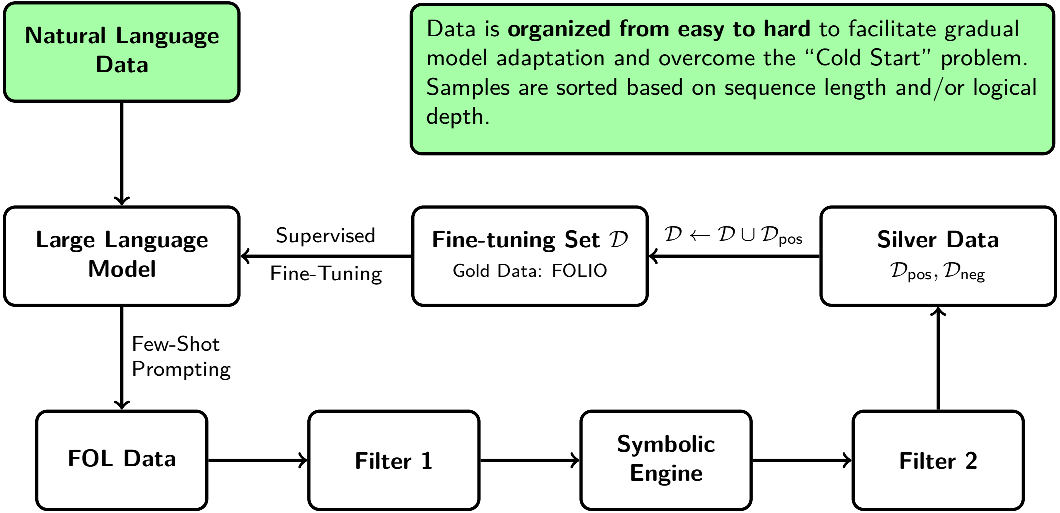


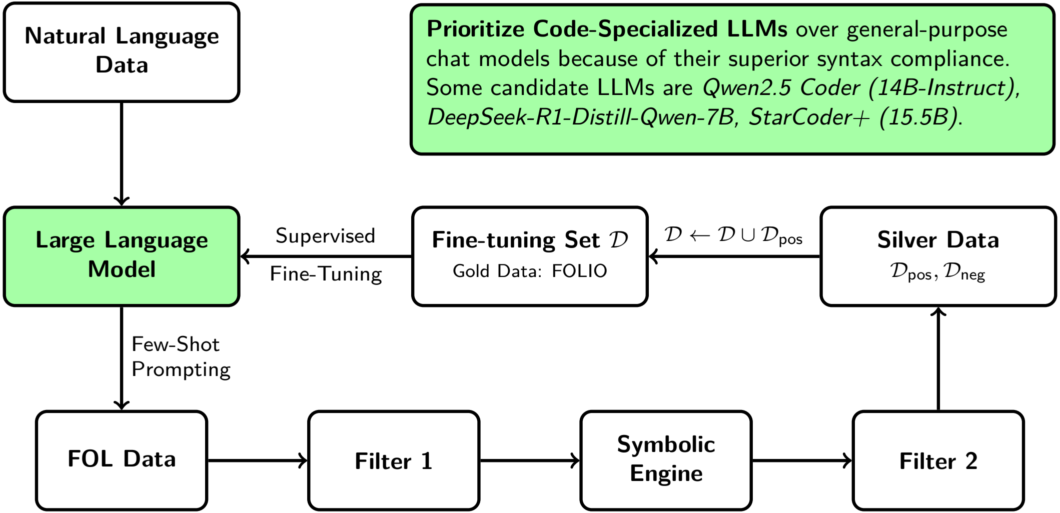
Data Pipeline

Data is **organized from easy to hard** to facilitate gradual model adaptation and overcome the "Cold Start" problem. Samples are sorted based on sequence length and/or logical depth.



Data Pipeline

Prioritize Code-Specialized LLMs over general-purpose chat models because of their superior syntax compliance. Some candidate LLMs are *Qwen2.5 Coder (14B-Instruct)*, *DeepSeek-R1-Distill-Qwen-7B*, *StarCoder+ (15.5B)*.



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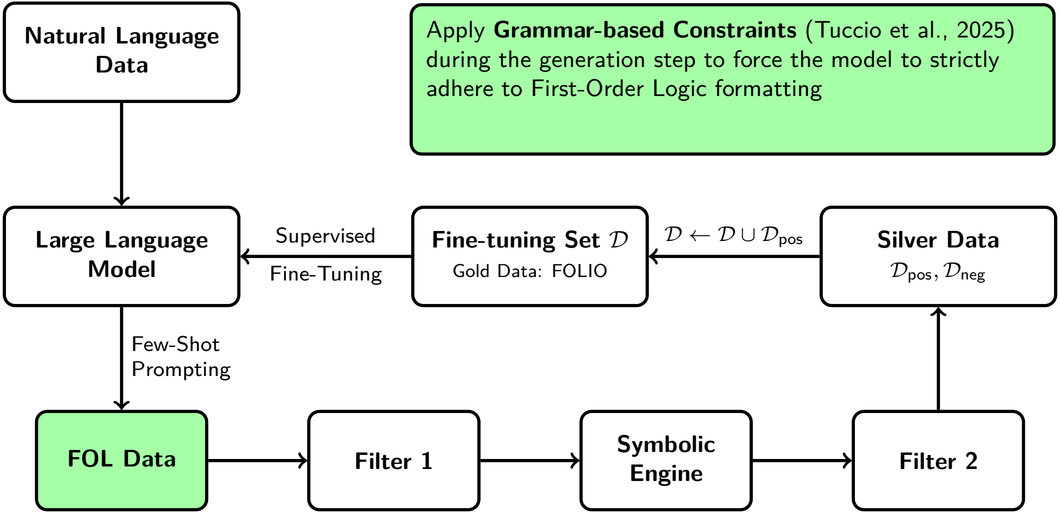
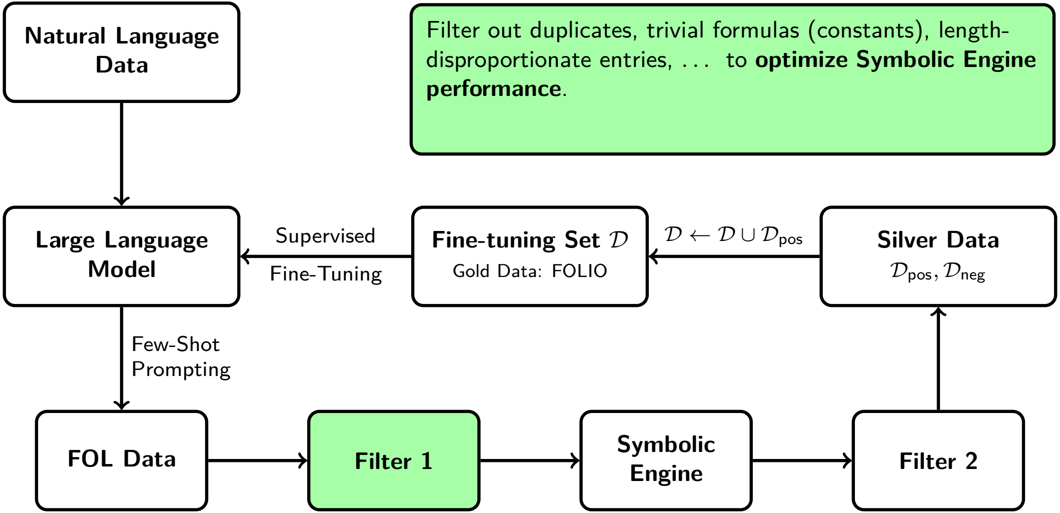


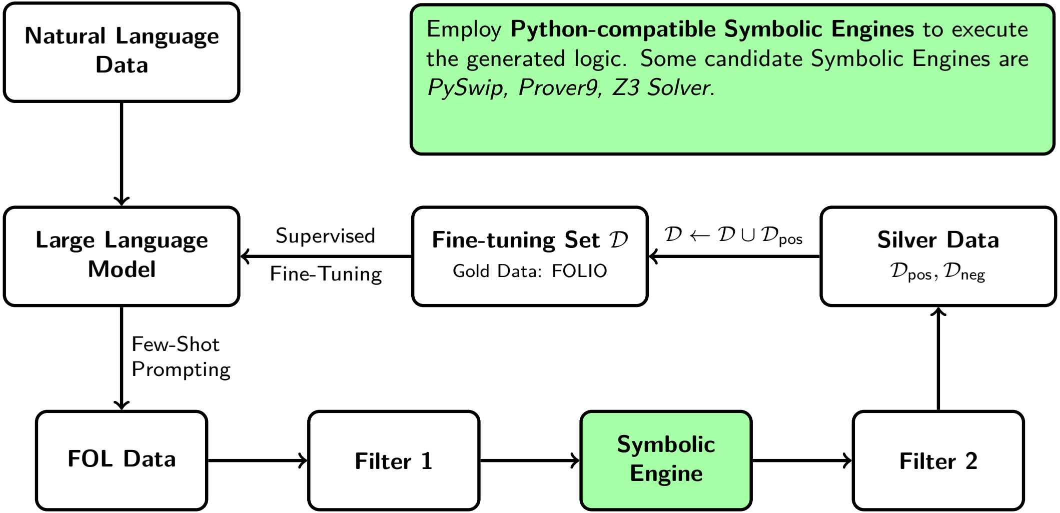
Data Pipeline

Filter out duplicates, trivial formulas (constants), length-disproportionate entries, . . . to **optimize Symbolic Engine performance**.



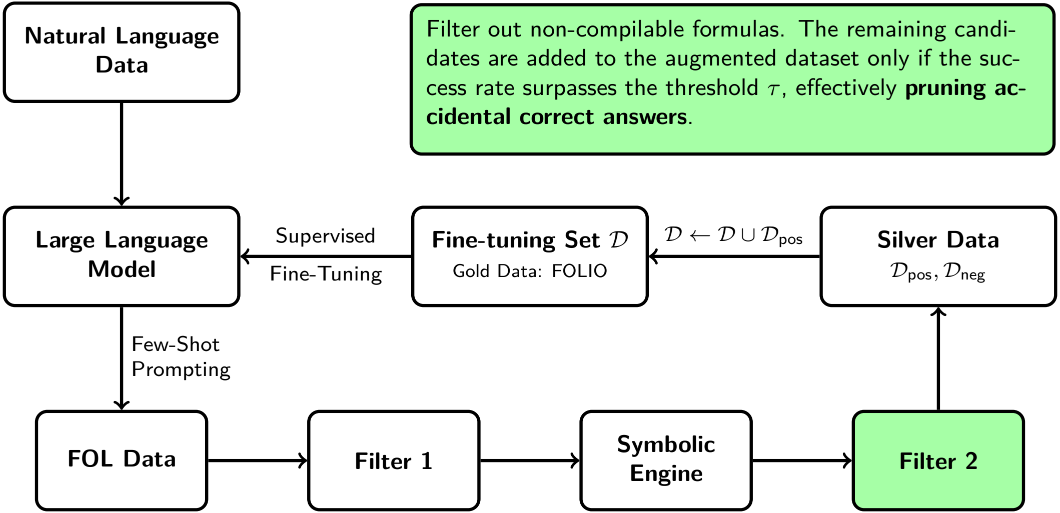
Data Pipeline

Employ **Python-compatible Symbolic Engines** to execute the generated logic. Some candidate Symbolic Engines are *PySwip, Prover9, Z3 Solver*.



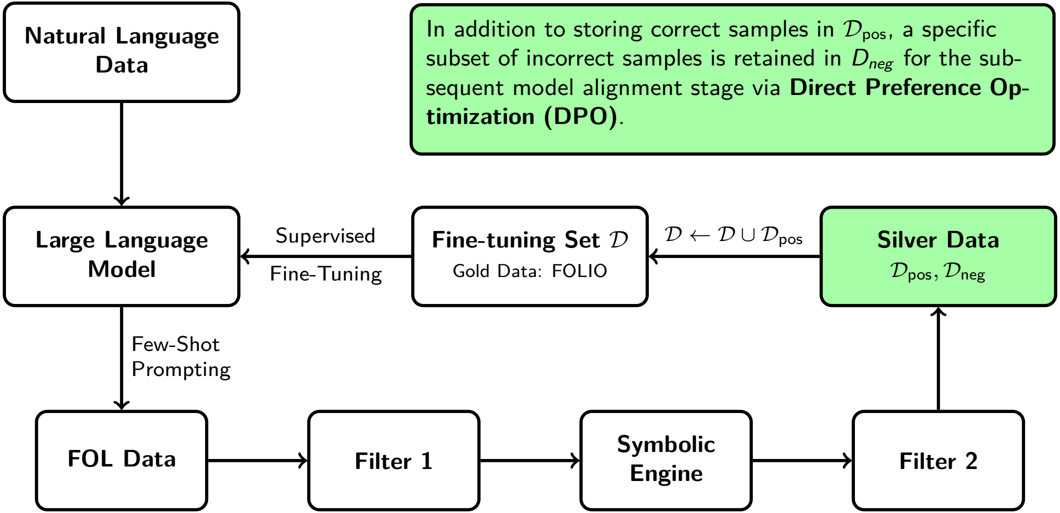
Data Pipeline

Filter out non-compilable formulas. The remaining candidates are added to the augmented dataset only if the success rate surpasses the threshold τ, effectively **pruning accidental correct answers**.



Data Pipeline

In addition to storing correct samples in $\mathcal{D}_{\text{pos}}$, a specific subset of incorrect samples is retained in D_{neg} for the subsequent model alignment stage via **Direct Preference Optimization (DPO)**.



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Related Work

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    - LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers (Olausson et al., EMNLP 2023)
    - LINA: An LLM-driven Neuro-Symbolic Approach for Faithful Logical Reasoning (Li et al., 2024)

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   - Absolute Zero: Reinforced Self-play Reasoning with Zero Data (Zhao et al., 2025)

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